AgroAI Doctor

AI-Based Crop Disease Detection System

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Project Overview

AgroAI Doctor is an AI-powered web application designed to detect crop diseases from leaf images. The system uses deep learning techniques to analyze leaf patterns and identify whether a plant is healthy or affected by a disease.

Users can upload a leaf image through a simple interface, and the system instantly provides results with a confidence score, disease details, and treatment suggestions. This helps users quickly understand the problem and take necessary action.

The platform aims to solve common agricultural challenges such as delayed diagnosis and lack of expert support. Since most farmers in Bangladesh are more comfortable communicating in Bangla, the system is fully designed in Bangla language and includes Bengali audio guidance. This makes the application more accessible, user-friendly, and effective for local users.

Overall, AgroAI Doctor combines artificial intelligence and web technology to provide a smart, efficient, and practical solution for crop disease detection.

Technology Used

Frontend: HTML, CSS, JavaScript

Backend: Lovable Platform

AI Model: TensorFlow / CNN

Other: Text-to-Speech (Bengali Audio)

Objectives

i. Detect crop diseases using AI
ii. Provide instant and accurate predictions
iii. Assist farmers with treatment suggestions
iv. Support Bengali language
v. Ensure user-friendly interface
vi. Reduce crop loss

Key Features

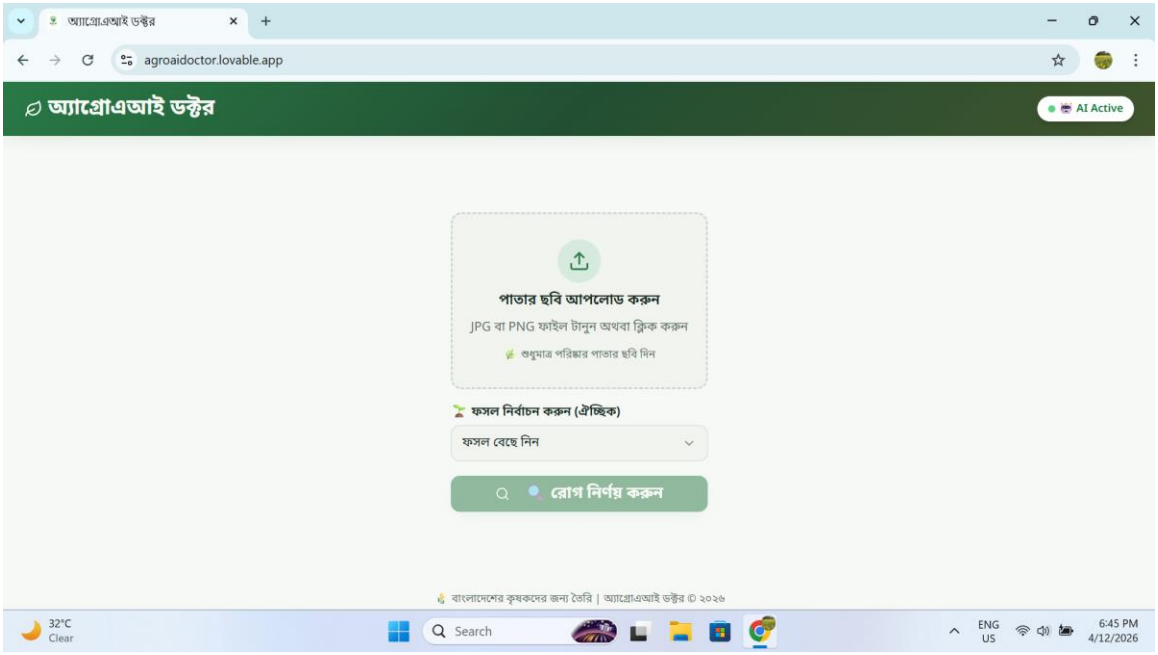
i. Leaf Image Upload
ii. AI-based Disease Detection
iii. Healthy vs Diseased Classification
iv. Confidence Score Display
v. Disease Details & Solutions
vi. Bengali Audio Explanation
vii. Invalid Image Detection

System Workflow

User → Upload Image → AI Processing → Disease Detection → Result → Solution → Audio Guidance

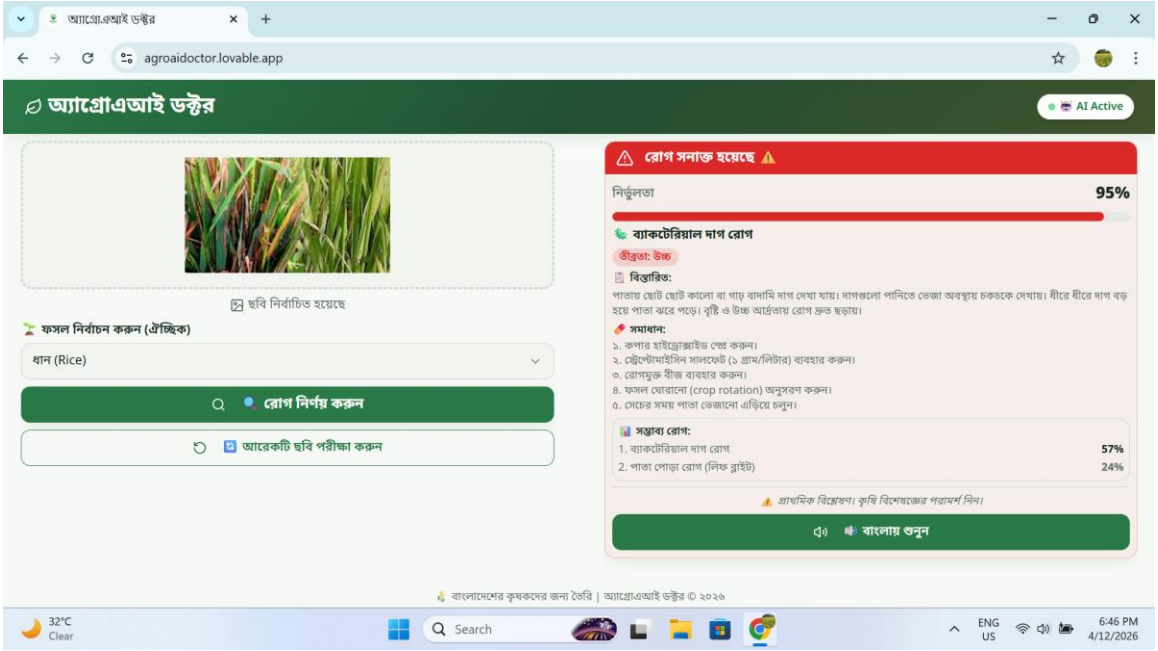
System Screenshots with Explanation

Figure 1: Upload Page - Users upload leaf images for analysis.



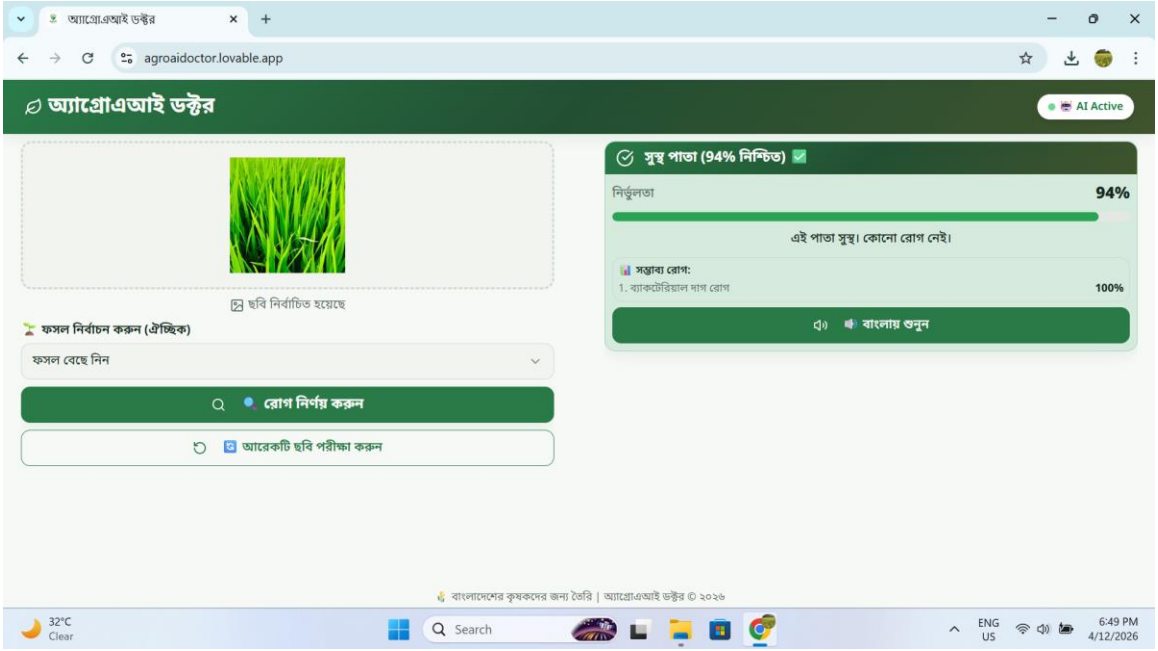
This interface allows users to upload leaf images (JPG/PNG format) for disease detection. The system provides clear instructions to ensure that only clean and focused leaf images are used for accurate prediction.

Figure 2: Disease Detection - Shows disease, severity, and solutions.



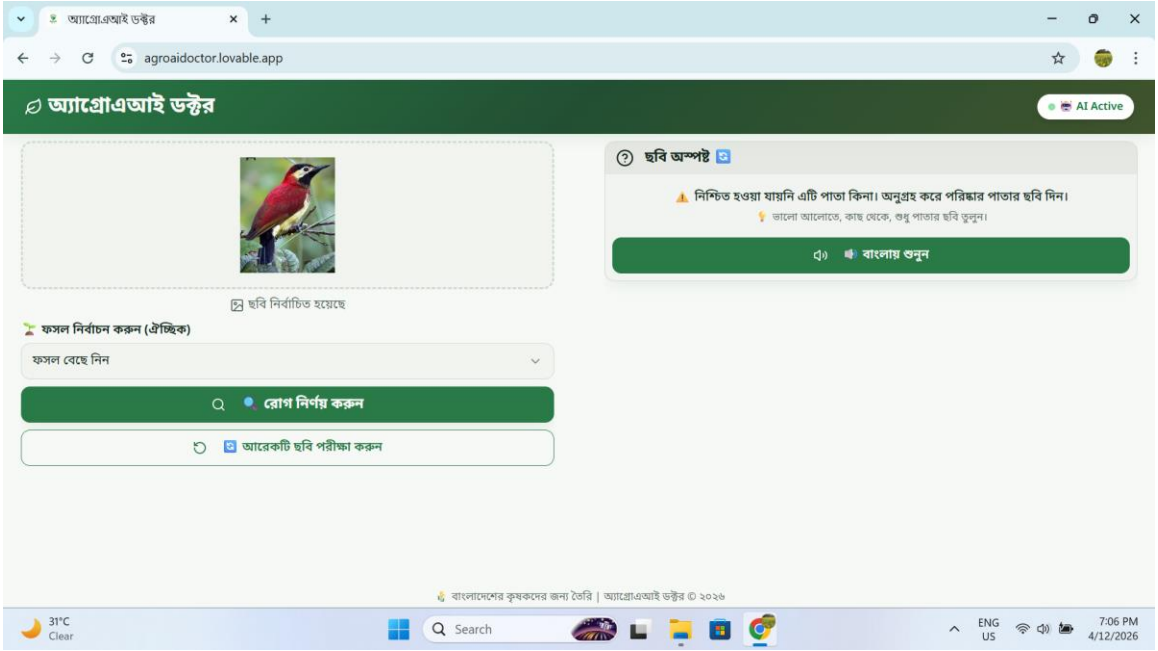
This figure shows the detection of a diseased leaf. The system identifies the disease (e.g., bacterial leaf blight), provides a confidence score, explains the symptoms, and suggests appropriate treatment methods.

Figure 3: Healthy Leaf - System confirms healthy leaf with confidence.



In this case, the system correctly identifies a healthy leaf and displays a high confidence score. It confirms that no disease is present, ensuring reliability and trust in the prediction.

Figure 4: Invalid Image - Non-leaf images are rejected with warning.



This figure demonstrates the system's ability to detect non-leaf images. When an invalid image (such as an object or animal) is uploaded, the system shows a warning message and prevents incorrect predictions.

Conclusion

AgroAI Doctor is a practical AI solution for agriculture. It helps farmers detect crop diseases early, providing solutions and reducing losses. The system integrates AI, web technologies, and local language support for real-world impact.

Project Links:

Live Demo : https://agroaidoctor.lovable.app/